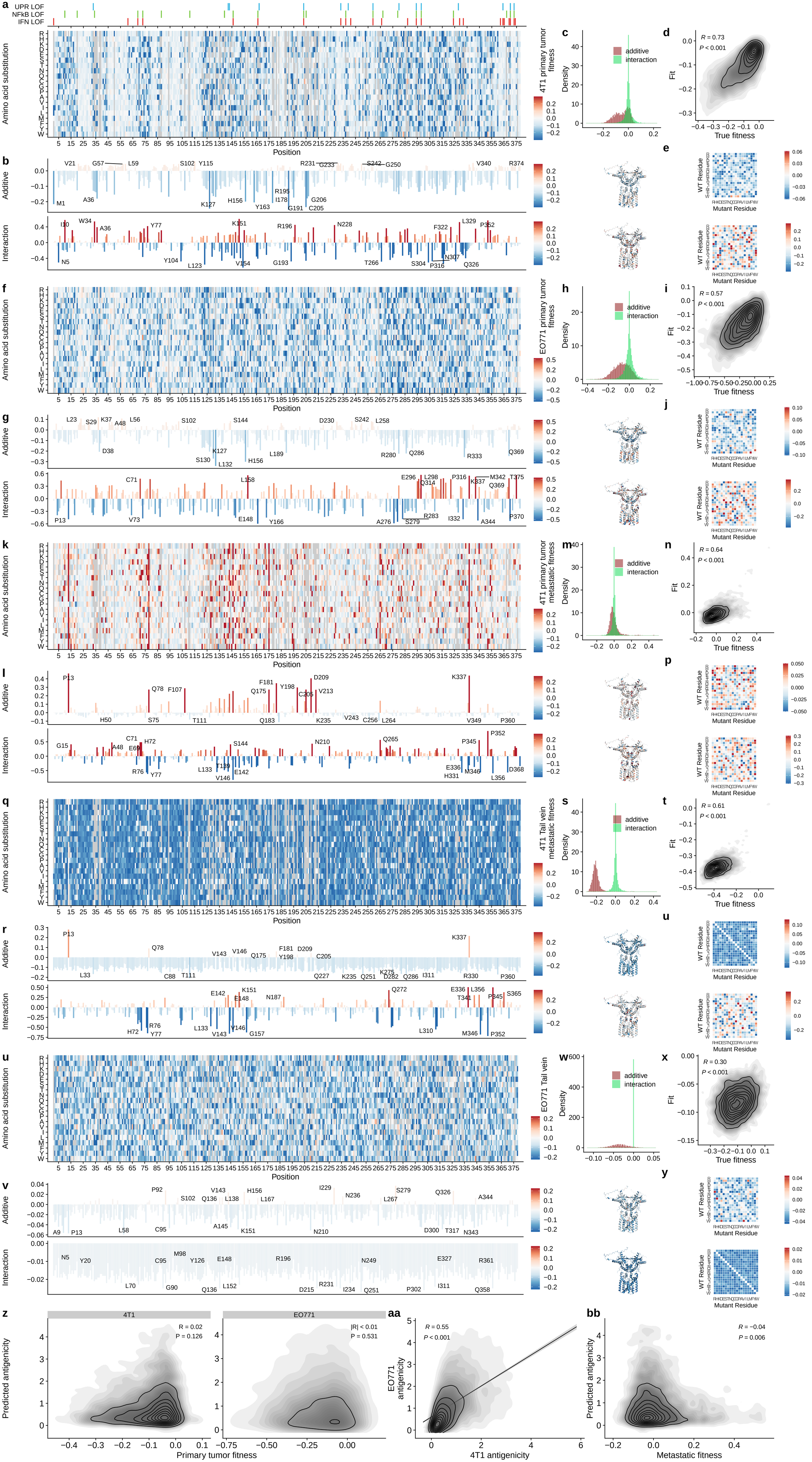

a
UPR LOF
NFkB LOF
IFN LOF
Amino acid substitution
Position
4T1 primary tumor fitness
Density
additive
interaction
Fit
True fitness
R = 0.73
P < 0.001
b
Additive
Interaction
WT Residue
Mutant Residue
EO771 primary tumor fitness
R = 0.57
P < 0.001
f
g
h
i
j
k
4T1 primary tumor metastatic fitness
R = 0.64
P < 0.001
l
m
n
p
q
4T1 Tail vein metastatic fitness
R = 0.61
P < 0.001
r
s
t
u
EO771 Tail vein
R = 0.30
P < 0.001
v
w
x
y
z
4T1
EO771
R = 0.02
P = 0.126
|R| < 0.01
P = 0.531
Predicted antigenicity
Primary tumor fitness
aa
EO771 antigenicity
4T1 antigenicity
R = 0.55
P < 0.001
bb
Metastatic fitness
R = −0.04
P = 0.006